

Overview

Supervised learning maps labeled inputs to outputs. Given dataset $D = \{(x_i, y_i)\}_{i=1}^n$, model learns function $f_{\text{heta}}(x)$ minimizing expected loss.

Two major task families: - **Regression:** target is continuous. - **Classification:** target is categorical.

Regression vs Classification

Regression

Predict real-valued quantity, e.g., house price. Typical losses: MSE, MAE.

Classification

Predict class label/probability, e.g., fraud vs non-fraud. Typical losses: log loss (cross-entropy).

Core Model Families

Linear Regression

Assumes linear relation:

$$\hat{y} = w^T x + b$$

Fast, interpretable, strong baseline for regression.

Logistic Regression

Models class probability:

$$P(y = 1|x) = \sigma(w^T x + b)$$

where σ is sigmoid. Despite name, used for classification.

k-Nearest Neighbors

Prediction from nearby samples in feature space. Non-parametric, sensitive to scaling and high dimensions.

Decision Trees

Recursive feature splits minimizing impurity (classification) or variance (regression). Interpretable but can overfit.

Ensemble Methods

Bagging

Train multiple base models on bootstrapped samples and aggregate predictions. Reduces variance.

Random Forests

Bagging of decision trees + random feature subsets per split. Usually strong default for tabular data.

Boosting

Sequentially add weak learners to correct predecessor errors. Reduces bias and can model complex patterns.

Gradient Boosting

Builds additive model by fitting learners to negative gradients of loss function.

XGBoost

Optimized gradient boosting with regularization, shrinkage, and system-level efficiency (parallelism, sparsity-aware handling).

Model Selection Tradeoffs

- Linear models: fast, interpretable, may underfit non-linear patterns.
- Random forests: robust, less tuning, larger memory.
- Gradient boosting/XGBoost: high performance on tabular data, careful tuning needed.
- k-NN: simple, expensive inference at scale.

Business Case Studies & Exceptions

Case 1: Credit Scoring Classification

Regulatory need for interpretability favored logistic regression baseline. Ensemble models improved AUC but required explainability overlays.

Case 2: House Price Regression

Linear model underfit neighborhood interactions; gradient boosting captured non-linearities and reduced error.

Case 3: High-Cardinality Features

Tree ensembles overfit sparse IDs without proper encoding and regularization.

Interview Questions & Answers

1. **Q: Define supervised learning.**
A: Learning mapping from labeled input-output pairs to generalize on unseen inputs.
2. **Q: Regression vs classification?**
A: Regression predicts continuous values; classification predicts categories/probabilities.
3. **Q: What is bagging?**
A: Bootstrap aggregating: train models on resampled data and aggregate outputs.
4. **Q: What is boosting?**
A: Sequentially train models where each focuses on previous errors.
5. **Q: Random Forest vs Gradient Boosting?**
A: RF reduces variance via independent trees; GB reduces bias via sequential corrections.
6. **Q: Why XGBoost often strong on tabular data?**
A: Regularized boosting, efficient implementation, and flexible objective/feature handling.
7. **Q: When prefer linear model over ensemble?**
A: Need interpretability, low latency, small data, or simple relationships.
8. **Q: Main risk with boosting?**
A: Overfitting and sensitivity to hyperparameters if unchecked.

9. **Q: Why feature scaling matters for k-NN/logistic?**
A: Distance/gradient behavior depends on feature magnitudes.
10. **Q: Baseline role in supervised projects?**
A: Establish reference performance before complex models.